

2.2.2 Examples

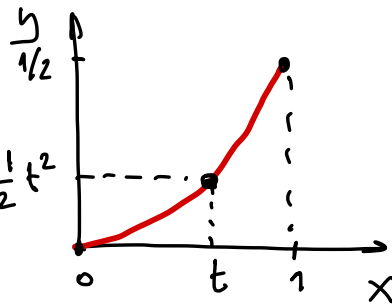
• Example 1: Compute $\int_{\Gamma} f \, dl$ for

$$a) \quad f: \mathbb{R}^2 \rightarrow \mathbb{R} \\ (x, y) \mapsto f(x, y) = \sqrt{x^2 + 4y^2}$$

and the curve Γ is param. as:

$$\gamma: [0, 1] \rightarrow \mathbb{R}^2$$

$$t \mapsto \gamma(t) = (t, \frac{1}{2}t^2)$$



$$\int_{\Gamma} f \, dl = \int_0^1 f(\gamma(t)) \|\gamma'(t)\| \, dt$$

$$\gamma'(t) = (1, t), \quad \|\gamma'(t)\| = \sqrt{1+t^2}$$

$$f(\gamma(t)) = f(\gamma_x(t), \gamma_y(t)) = *$$

$$\gamma(t) = \begin{pmatrix} \gamma_x(t) = t \\ \gamma_y(t) = \frac{1}{2}t^2 \end{pmatrix}$$

$$(*) = \sqrt{\gamma_x^2 + 4\gamma_y^2} = \sqrt{t^2 + 4\left(\frac{1}{2}t^2\right)^2} = \sqrt{t^2 + t^4}$$

$$\int_{\Gamma} f \, d\ell = \int_0^1 \underbrace{\sqrt{t^2 + t^4}}_{t\sqrt{1+t^2}} \sqrt{1+t^2} \, dt = \int_0^1 t(1+t^2) \, dt \stackrel{\text{DIY}}{=} \dots = \frac{3}{4}$$

b) $f: \mathbb{R}^2 \rightarrow \mathbb{R}$

$$(x, y) \mapsto f(x, y) = x$$

and the curve $\Gamma = \{(x, y) \in \mathbb{R}^2 : y = \cosh x \text{ for } x \in [0, 1]\}$.

We find the parameterization:

$$\gamma: [0, 1] \rightarrow \mathbb{R}^2$$

$$t \mapsto \gamma(t) = (t, \cosh t)$$

$$\gamma'(t) = (1, \sinh t), \quad \|\gamma'(t)\| = \sqrt{\underbrace{1 + \sinh^2 t}_{\cosh^2 t}} = \cosh t$$

$$f(\gamma(t)) = t.$$

$$\int_{\gamma} f \, d\ell = \int_0^1 t \cosh t \, dt = \overset{\text{by parts}}{\underset{\substack{\text{DIY} \\ \text{(H3)}}}{-}} = \sinh(1) - \cosh(1) + 1$$

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$$= \frac{\cancel{e^1} - e^{-1}}{2} - \frac{\cancel{e^1} + e^{-1}}{2} + 1 = 1 - e^{-1}$$

• Example 2:

$$a) F: \mathbb{R}^3 \rightarrow \mathbb{R}^3$$

$$(x, y, z) \mapsto F(x, y, z) = (x, z, y)$$

$$\gamma: [0, 2\pi] \rightarrow \mathbb{R}^3$$

$$t \mapsto \gamma(t) = (\cos t, \sin t, 0)$$

$$\int_{\gamma} F \circ dl = \int_0^{2\pi} F(\gamma(t)) \cdot \gamma'(t) dt$$

$$\gamma'(t) = (-\sin t, \cos t, 0)$$

$$F(\gamma(t)) = F(\gamma_x(t), \gamma_y(t), \gamma_z(t)) = (\gamma_x(t), \gamma_z(t), \gamma_y(t))$$

$$= (\cos t, 0, \sin t)$$

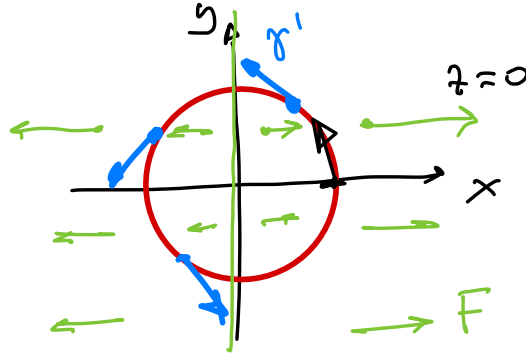
$$\int_{\Gamma} \mathbf{F} \cdot d\mathbf{l} = \int_0^{2\pi} (\cos t, 0, \sin t) \cdot (-\sin t, \cos t, 0) dt$$

$$= - \int_0^{2\pi} \cos t \sin t dt = \frac{Dy}{(T5)} = 0$$

or (T1) + standard integration

Note:

$$\mathbf{F}(x, y, z) = (x, z, y)$$



$$\int_{\Gamma} \mathbf{F} \cdot d\mathbf{l} = \int_0^{2\pi} \mathbf{F}(\gamma(t)) \cdot \gamma'(t) dt$$

$$b) \quad F: \mathbb{R}^3 \rightarrow \mathbb{R}^3$$

$$(x, y, z) \mapsto F(x, y, z) = (x^2, y^3, z^2)$$

and the curve $\Gamma = \{ (x, y, z) \in \mathbb{R}^3 : y = e^x \text{ and } z = x \text{ for } x \in [0, 1] \}$.

$$\gamma(t) : [0, 1] \rightarrow \mathbb{R}^3$$

$$t \mapsto (t, e^t, t)$$

Home exercise:

$$\gamma(t) : [\quad] \rightarrow \mathbb{R}^3$$

$$t \mapsto (\log t, t, \log t)$$

$$\int_{\Gamma} F \cdot d\ell = \int_0^1 F(\gamma(t)) \cdot \gamma'(t) dt$$

$$\gamma'(t) = (1, e^t, 1)$$

$$F(\gamma(t)) = (t^2, e^{3t}, t^2)$$

$$\begin{aligned} \int_{\gamma} F \cdot d\mathbf{l} &= \int_0^1 (t^2, e^{3t}, t^2) \cdot (1, e^t, 1) dt \\ &= \int_0^1 (2t^2 + e^{4t}) dt = \text{Div} = \frac{5 + 3e^4}{12} \end{aligned}$$

Example 3: compute the length of Γ , Γ is the circle of radius R and centered at the origin.

$$\text{length}(\Gamma) = 2\pi R$$

$$\gamma: [0, 2\pi] \longrightarrow \mathbb{R}^2$$

$$t \longmapsto \gamma(t) = R(\cos t, \sin t)$$

Note:

$$\gamma: [0, 1] \rightarrow \mathbb{R}^3$$

$$t \mapsto \gamma(t) = R (\cos 2\pi t, \sin 2\pi t)$$

$$\text{length } \gamma = \int_{\gamma} 1 \, d\ell = \int_0^{2\pi} \underbrace{1(\gamma(t))}_1 \| \gamma'(t) \| \, dt = *$$

$$\gamma'(t) = R (-\sin t, \cos t), \quad \| \gamma'(t) \| = R$$

$$* = \int_0^{2\pi} 1 \cdot R \, dt = R \int_0^{2\pi} dt = 2\pi R.$$